

CS 201 CS 201: Class Notes

Comprehensive Lecture Notes & Course Companion

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LECTURE 1

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Introduction to Algorithm Analysis

Welcome to CS 201! In this first lecture, we set the foundation for computational complexity analysis, mathematical modeling of program performance, and asymptotic notation.

[DEF] Definition 1 (Algorithm)

An **algorithm** is a finite, well-defined sequence of step-by-step computational instructions that transforms an input into a desired output.

When analyzing algorithms, we evaluate performance across two main resource axes:

1. **Time Complexity:** How the execution runtime scales as the input size n approaches infinity.
2. **Space Complexity:** Memory auxiliary consumption required by the data structures during execution.

[THM] Theorem 1 (Master Theorem for Divide-and-Conquer)

Let $T(n)$ be a recurrence relation defined by $T(n) = aT(n/b) + f(n)$ where $a \geq 1$ and $b > 1$. If $f(n) = \Theta(n^c)$, then:

$$T(n) = \begin{cases} \Theta(n^{\log_b a}) & \text{if } c < \log_b a \\ \Theta(n^c \log n) & \text{if } c = \log_b a \\ \Theta(n^c) & \text{if } c > \log_b a \end{cases}$$

Proof: By constructing a recurrence tree of depth $\log_b n$, at level i there are a^i subproblems, each of size n/b^i . Summing work across all levels yields the geometric series solution above. ■

[EX] Example 1 (Binary Search Analysis)

In Binary Search, $a = 1$, $b = 2$, and $f(n) = \Theta(1)$. Since $c = 0$ and $\log_2 1 = 0$, Case 2 of the Master Theorem applies:

$$T(n) = \Theta(\log n)$$

[!] Warning

Master Theorem does not apply if $f(n)$ is not polynomial (e.g., $f(n) = 2^n$) or if $a < 1$.

[KEY] Key Takeaways

- Big-O provides an asymptotic upper bound on growth rate.
- Master Theorem simplifies divide-and-conquer recurrences.
- Always consider both average-case and worst-case bounds.

LECTURE 2

September 7, 2026

Dynamic Programming & Optimization

Dynamic Programming (DP) is an algorithmic technique for solving optimization problems by breaking them down into simpler overlapping subproblems.

[DEF] Definition 2 (Optimal Substructure)

A problem exhibits **optimal substructure** if an optimal solution to the problem contains optimal solutions to its subproblems.

Python Implementation: 0/1 Knapsack Bottom-Up

```
1 def knapsack(weights, values, capacity):
2     n = len(weights)
3     dp = [[0] * (capacity + 1) for _ in range(n + 1)]
4
5     for i in range(1, n + 1):
6         for w in range(1, capacity + 1):
7             if weights[i-1] <= w:
8                 dp[i][w] = max(values[i-1] + dp[i-1][w - weights[i-1]], dp[i-1][w])
9             else:
10                dp[i][w] = dp[i-1][w]
11
12     return dp[n][capacity]
```

[*] Pro Tip

When designing DP state transitions, start by writing down the mathematical recurrence relation before touching code.

[EXR] Exercise 1 (Fibonacci DP Space Optimization)

Reduce the space complexity of bottom-up Fibonacci calculation from $O(n)$ to $O(1)$.

Solution:

Instead of storing an array of size n , track only the last two state variables (a and b) and update them iteratively in a loop:

$$(a, b) \leftarrow (b, a + b)$$